Deep Learning vs. Conventional Methods for Tabular Data Imputation

immediate

Abstract

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- Enough background content to give context to the study AND
- A brief statement of primary results AND
- A short concluding sentence.

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1 Introduction

Your manuscript should contain all of the numbered sections specified in this template: Introduction, Materials and Methods, Results, Discussion. Please do not reorder or retitle the section headings.

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For example, see Equation 1 and Equation 2 below.

$$a^2 + b^2 = c^2 \tag{1}$$

$$\begin{aligned} A &= \frac{\pi r^2}{2} \\ &= \frac{1}{2}\pi r^2 \end{aligned} \tag{2}$$

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2 Methods

2.1 Literature Search Procedure

Begin with a section titled Experimental Design describing the objectives and design of the study as well as prespecified components.

2.2 Conventional Statistical Methods

2.2.1 Deletion Methods

2.2.1.1 Listwise Deletion Listwise Deletion, also known as Complete Case Analysis (CCA), is the most direct method for handling missing values. It removes an entire row from the dataset if it contains one or more missing values. The main advantage of this approach is that it is extremely simple to implement and requires no additional computation. However, this method may substantially reduce the sample size and consequently decrease statistical power.[A review on]

2.2.1.2 Pairwise Deletion Pairwise Deletion, also known as Available Case Analysis (ACA), is a method used to handle missing values. It excludes observations only when the analysis involves variables with missing data, rather than removing the entire case. The main advantage of this approach is that it retains more available data and preserves a larger sample size and statistical power. However, this method is appropriate only under the MCAR assumption. Different analyses may use different sample sizes. This can make the correlation matrix difficult to interpret and may produce a non-positive definite matrix, which can affect subsequent model estimation.

2.2.2 Mean / Median Imputation

Mean / Median Imputation is one of the most common single imputation methods in statistics. It replaces missing values with the mean or median of the observed values for that variable. This method is simple to implement and computationally efficient. It can also provide a practical solution when the dataset is very small. However, this method is appropriate only when the missing data mechanism satisfies MCAR. It may underestimate the variance and standard errors and bias the correlations among variables. In addition, the mean is sensitive to extreme values and cannot capture complex nonlinear relationships.

2.2.3 MICE

2.3 Machine Learning Methods

2.3.1 K-Nearest Neighbors (KNN)

KNN imputation is a non-parametric method because it does not assume a specific functional form or probability distribution for the data. Instead, missing values are estimated using the observed values of the K most similar instances in the dataset.[Deep learning]

This method estimates missing values using the K most similar observations in the dataset. First, the distance between the incomplete observation and all other observations is computed using a distance metric such as Euclidean distance. Next, the K nearest neighbors are identified. For numerical variables, the missing value is imputed using the mean or weighted mean of the neighbors. If the missing value is a category, the algorithm uses the mode (the most frequent value) among the neighbors.[How To Treat]

The KNN imputation method offers several advantages over traditional missing data handling techniques such as mean imputation. First, KNN preserves the underlying relationships among variables because the imputed values are derived from observations with similar feature patterns rather than from global summary statistics. Second, KNN is a non-parametric method and therefore does not require assumptions about the underlying data distribution. This makes the method flexible and applicable to a wide range of datasets. Third, KNN can accommodate both continuous and categorical variables by using appropriate aggregation strategies such as the mean for numerical attributes and the mode for categorical attributes. Finally, because KNN relies on similarity among observations, it is capable of capturing local structures in the data, which can lead to more accurate imputations compared with simple statistical approaches.

Despite its advantages, the KNN imputation method also has several limitations. One significant issue is its computational cost. Because the algorithm must search through the entire dataset to identify the nearest neighbors for each missing value, it can become computationally expensive and memory-intensive when applied to very large datasets. Another challenge lies in its sensitivity to hyperparameters. The effectiveness of the method depends heavily on the choice of the parameter k and the quality of the features used to compute distances. If the selected features are weakly related to the variable with missing values, the imputation results may be inaccurate.[AStudyOn]

2.3.2 MissForest

MissForest is a non-parametric imputation method that utilizes Random Forest to estimate missing values. The algorithm first initializes missing entries using simple statistical methods, such as mean imputation for continuous variables and mode imputation for categorical variables. Variables are then ordered according to the amount of missing data, starting with those with the fewest missing values. For each variable with missing data, a Random Forest model is trained using the observed values as the response variable and the remaining variables as predictors. The trained model is then used to predict the missing values. This process is repeated iteratively, and the imputed dataset is updated in each step. The algorithm stops when the difference between successive imputations begins to increase.

The main strength of MissForest is its ability to capture non-linear relationships and complex interactions between variables. Unlike methods such as KNN and MICE, MissForest can handle both continuous and categorical variables directly. It not only avoids separate processing for different data types but also eliminates the need for additional transformations, such as standardization or dummy coding. Another advantage is that it does not require strong assumptions about the data distribution. Instead, it uses random forest models to learn relationships between variables directly from the data. In addition, MissForest can estimate imputation quality using the Out-of-Bag (OOB) error, so a separate test dataset is not required.

Although MissForest has several advantages, it also has some limitations. Since the algorithm repeatedly trains random forest models during the iterative imputation process, it may require more computational time when applied to large datasets. Instead, a more relevant issue for small datasets is the potential risk of overfitting. Because MissForest relies on random forest models, the algorithm may learn patterns that are overly specific to the limited sample. Therefore, when applying MissForest to small-sample survey data, it is important to carefully examine the stability and plausibility of the imputed values.

Empirical studies suggest that MissForest performs particularly well when the dataset contains more than 1,000 observations, often achieving better imputation accuracy than traditional methods such as mean imputation or KNN-based approaches. [A review on missing] In addition, prior research indicates that for tabular datasets with moderate sample sizes (typically fewer than 30,000 observations), MissForest may outperform deep learning approaches such as GAIN or variational autoencoders (VAE). Therefore, for survey datasets with sample sizes between approximately 1,000 and 30,000 observations, MissForest can be considered a suitable method for handling missing values.[Versus]

2.4 Deep Learning Methods

The integration of deep learning into the field of data imputation primarily involves two frameworks: Generative Adversarial Networks (GANs) and Transformer-based models (Attention mechanisms).

2.4.1 Generative Frameworks

The core objective of generative frameworks is to learn the joint probability distribution of the data. Rather than simply predicting a value based on other features, these models attempt to reconstruct the underlying data-generating logic to produce imputations that maintain global statistical characteristics.

- **Generative Adversarial Networks (GANs):** These consist of a Generator, which observes available data to predict missing values, and a Discriminator, which attempts to distinguish between observed and imputed data points. GAN-based models exhibit higher resilience under high missingness rates (e.g., above 50%) because they capture the overall data generation logic rather than local correlations. GAIN [**yoon2018gain**] serves as a pivotal benchmark in this field, introducing a "hint vector" to guide the discriminator and force the generator to learn the true underlying distribution.
- **Variational Autoencoders (VAEs):** VAEs compress incomplete data into a low-dimensional latent space to learn the probability distribution before decoding and reconstructing the complete data matrix. A significant advantage of VAEs is their ability to provide uncertainty quantification, which is critical for risk-sensitive scenarios such as healthcare. MIWAE [**mattei2019miwae**] optimized importance sampling for incomplete datasets, while TVAE was specifically designed to handle tabular heterogeneity.
- **Diffusion Models:** These models simulate physical diffusion processes to reconstruct missing values through iterative denoising. TabDDPM [**kotelnikov2023tabddpm**] has been shown in recent 2025 literature to outperform GANs and VAEs in preserving original data distributions (e.g., minimizing KL divergence). The latest MissDDIM [**zhou2025missddim**] further addresses the inference latency of diffusion models by replacing stochastic sampling with a deterministic non-Markovian sampling path, achieving faster and more consistent imputation results.

2.4.2 Attention-based Frameworks

The essence of attention mechanisms lies in dynamic weight allocation. When imputing a specific missing cell, the model automatically determines which other features or similar samples are most informative for the current estimation.

- **TabNet:** This architecture utilizes sequential attention to "mask" features at each decision step, mimicking the splitting logic of decision trees. By employing the Sparsemax function to generate sparse weights, TabNet [**arik2021tabnet**] achieves tree-like interpretability while effectively modeling complex non-linear interactions between features.

- **Transformer Variants:** Through self-attention mechanisms, these models capture not only "between-feature" correlations within a single row but also "between-sample" relationships across different rows. FT-Transformer [**gorishniy2021revisiting**] transforms features into embeddings for processing, while the Non-Parametric Transformer (NPT) [**kossen2021self**] allows the model to communicate across all samples in a dataset. Recent research has further optimized these frameworks to enhance fault tolerance and computational efficiency.

- **Tabular Foundation Models and In-Context Learning (ICL):** Frameworks such as TabPFN eliminate the need for retraining on missing data. Instead, they treat the training set as "context" for the Transformer, leveraging pre-trained weights to predict missing values in the test set via a single forward pass. TabImpute [**feitelberg2025tabimpute**] introduced entry-wise featurization, achieving a 100x speedup over traditional TabPFN imputation methods and setting a new benchmark for zero-shot data imputation.

3 Results

The results should describe the experiments performed and the findings observed. The results section should be divided into subsections to delineate different experimental themes.

- All data should be presented in the Results. No data should be presented for the first time in the Discussion. Data (such as from Western blots) should be appropriately quantified.

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4 Discussion

In this section, we will introduce the experiments conducted to evaluate the differences between traditional machine learning and deep learning methods for data imputation.We designed a two-phase experimental approach: Phase 1 compares the data recovery accuracy of different imputation methods, and Phase 2 evaluates the prediction accuracy for gender classification after imputation.

4.1 Experimental Setup

4.1.1 Environment and Tools

All experiments were conducted on the Kaggle online platform to ensure a consistent computing environment, with a fixed random seed of 42 applied to all data splitting and model initialization processes to ensure reproducibility.

4.1.2 Dataset Descriptions

To evaluate the impact of sample size on different imputation techniques, we selected two datasets with significantly different scales:

- **Small-scale Dataset (Boy or Girl):** This dataset contains approximately 400 samples, representing a typical small-sample scenario. It includes various features such as height, weight, star sign, phone OS, and IQ. The target variable is gender, which is used to evaluate the performance of imputation methods on downstream classification tasks.

- **Large-scale Dataset (CDC BRFSS):** Downloaded via the `kagglehub` library, this dataset contains hundreds of thousands of health-related records from the Behavioral Risk Factor Surveillance System (BRFSS). This large-scale dataset is used to assess whether deep learning models demonstrate superior representation capabilities when a vast amount of data is available.

4.1.3 Experimental Design and Protocol

We use a control-variable approach to ensure that all processes remain identical except for the choice of the imputation method. The experimental pipeline is organized as follows:

- **Data Preprocessing and Cleaning:** The raw datasets undergo initial cleaning to handle anomalies. Categorical features are transformed using *Label Encoding* to ensure they are compatible with both machine learning and deep learning algorithms.

- **Phase 1: Recovery Accuracy Analysis:** To test the reconstructive capability of each method, we simulate data loss by randomly masking 10% of the values in the continuous features. Various imputation techniques are then applied to fill these gaps, and their performance is evaluated using distance-based metrics (such as MAE and R^2) by comparing the imputed values against the original ground truth.

- **Phase 2: Downstream Performance Evaluation:** This phase investigates the impact of different imputation methods on the final predictive outcome. Using a unified downstream classification model, we measure the gender prediction accuracy. This allows us to determine which imputation strategy provides the highest utility for practical classification tasks.

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General

Thank others for any contributions, whether it be direct technical help or indirect assistance

Author Contributions

Describe contributions of each author to the paper, using the first initial and full last name.

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"S. Zhang conceived the idea and designed the experiments."

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"All authors contributed equally to the writing of the manuscript."

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